

Lab Assignment 03

Intermediate Biomedical Data Science (EN.580.375)

Overview

This lab practices unsupervised learning from Lectures 6–7: dimensionality reduction, similarity measures, and unsupervised clustering.

Groups are allowed, but each student must submit their own code and short write-up. If you work with others, identify your collaborators and write your responses in your own words. Include your code, plots, numerical results, and short explanations in the submitted notebook or equivalent file.

You may use Google Colab or a local Jupyter environment. The code chunks below are starter code: run and adapt them as needed, and include your code, plots, numerical results, and short explanations in the submitted notebook or equivalent file.

Suggested schedule

- **Monday:** Part 1 — PCA, explained variance, and scores/loadings
- **Wednesday:** Part 2 — similarity, heatmaps, hierarchical clustering, k -means clustering, graph-based clustering, and evaluation

Part 1 — Monday: Dimensionality reduction

1. Why reduce dimension?

Suppose a gene-expression study measures $p = 20,000$ genes in n samples.

Questions

1. Why can we not simply inspect every pairwise scatterplot? Write the number of possible plots as a function of p and evaluate it for $p = 20,000$.
2. Give two reasons why a visually separated group in a PCA plot is not automatically a biological cell type.

2. PCA: the direction with the largest variance

Use the following centered and scaled toy expression matrix. Rows are samples and columns are genes.

```
import numpy as np
import matplotlib.pyplot as plt
from sklearn.decomposition import PCA

rng = np.random.default_rng(375)
n_samples = 180
```

```

n_genes = 20

X = rng.normal(size=(n_samples, n_genes))
X[:90, :3] += 1.5
X[90:, :3] -= 1.5
X = (X - X.mean(axis=0)) / X.std(axis=0, ddof=1)

```

For a set of features $X_1, \dots, X_p$, the first principal component is the normalized linear combination

$$Z_1 = \phi_{11}X_1 + \phi_{21}X_2 + \dots + \phi_{p1}X_p, \quad \sum_{j=1}^p \phi_{j1}^2 = 1,$$

that has the largest variance.

Questions

1. Write the first principal component loading vector using the notation $\phi_1 = (\phi_{11}, \dots, \phi_{p1})^T$.
2. State the optimization problem for the first principal component using x_{ij} and z_{i1} .
3. In your own words, explain why the constraint $\sum_{j=1}^p \phi_{j1}^2 = 1$ is necessary.
4. What do the scores $z_{11}, \dots, z_{n1}$ represent? What do the loadings $\phi_{11}, \dots, \phi_{p1}$ represent?

3. PCA scores, loadings, and explained variance

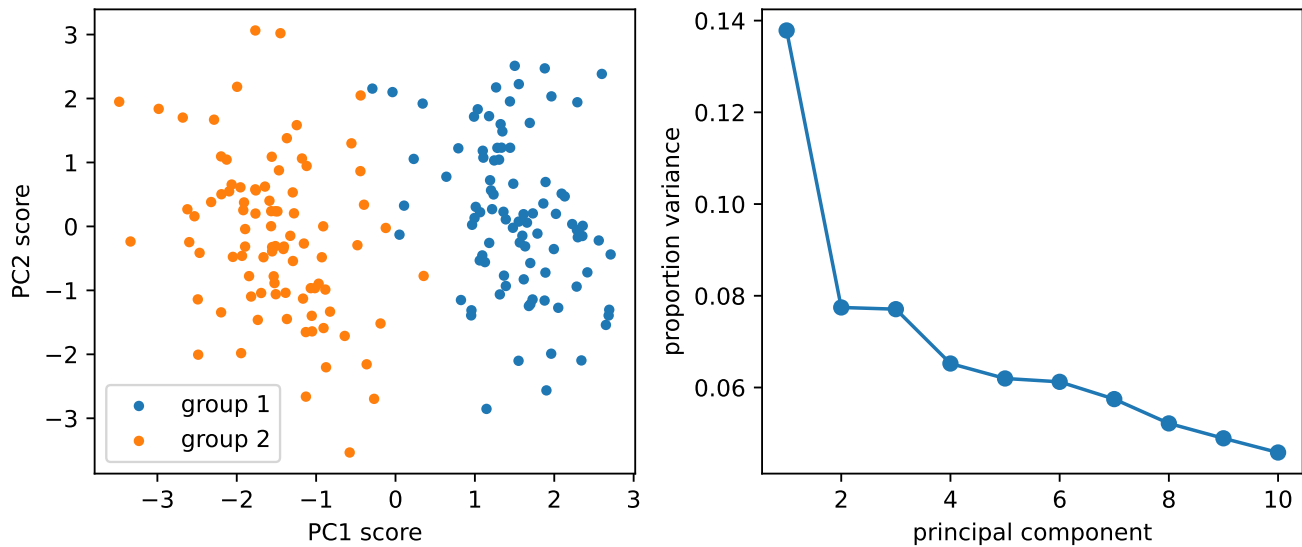
Fit PCA and make a scatterplot of the first two scores and a plot of the proportion of variance explained by each component.

```

pca = PCA(n_components=10).fit(X)
scores = pca.transform(X)

fig, ax = plt.subplots(1, 2, figsize=(8, 3.5))
ax[0].scatter(scores[:90, 0], scores[:90, 1], s=12, label="group 1")
ax[0].scatter(scores[90:, 0], scores[90:, 1], s=12, label="group 2")
ax[0].set(xlabel="PC1 score", ylabel="PC2 score")
ax[0].legend()
ax[1].plot(np.arange(1, 11), pca.explained_variance_ratio_, marker="o")
ax[1].set(xlabel="principal component", ylabel="proportion variance")
plt.tight_layout()

```



Questions

1. Which PC explains the most variation? How much is explained by the first two PCs, and how many would you retain to explain at least 80%?
2. Plot the PC1 loadings. Which genes contribute most? Compare PC1 scores with the known groups and name one check needed before calling the separation biological.

Part 2 — Wednesday: Similarity and clustering

5. Similarity and dissimilarity

Create a small set of samples with two expression features.

```
from scipy.spatial.distance import pdist, squareform

X_small = np.array([
    [1.0, 1.2], [1.4, 0.9], [0.8, 1.1],
    [4.0, 4.2], [4.4, 3.8], [3.7, 4.1],
])
sample_names = ["A", "B", "C", "D", "E", "F"]
D = squareform(pdist(X_small, metric="euclidean"))
```

Questions

1. What does D_{ij} mean, and why is $D_{ii} = 0$? Which pair is closest and which is farthest?
2. Repeat with Manhattan distance. Do the pairs change? Why is choosing a distance a scientific choice, and when might Euclidean distance mislead?

6. Heatmaps and hierarchical clustering

Display the pairwise distance matrix as a heatmap and use agglomerative hierarchical clustering to reorder the samples.

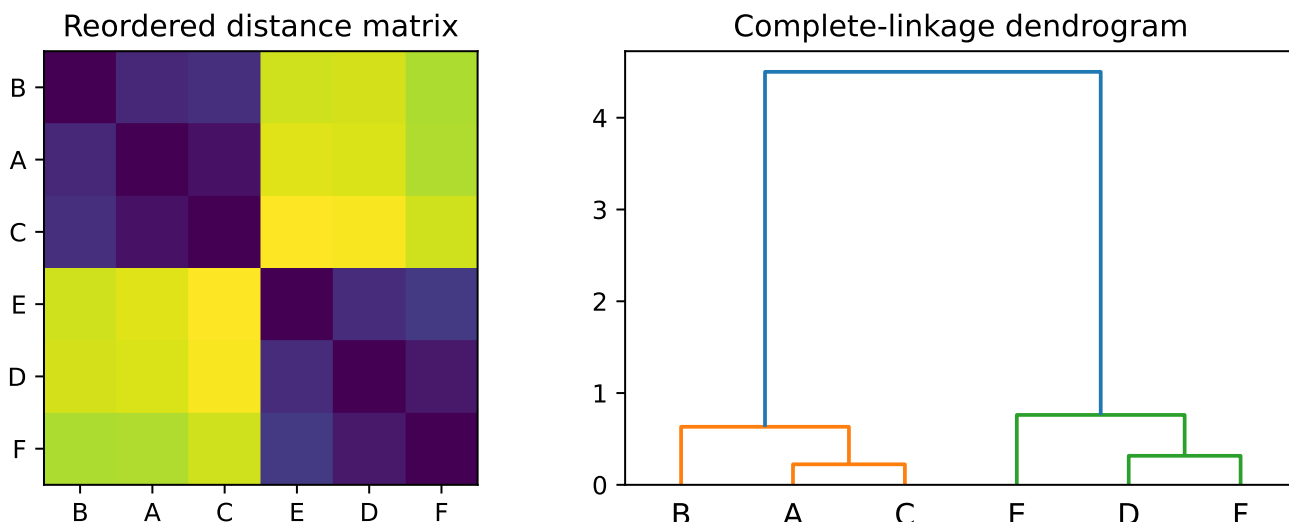
```
from scipy.cluster.hierarchy import linkage, dendrogram, fcluster
```

```

Z_link = linkage(X_small, method="complete", metric="euclidean")
order = dendrogram(Z_link, labels=sample_names, no_plot=True)["leaves"]

fig, ax = plt.subplots(1, 2, figsize=(8, 3.2))
ax[0].imshow(D[np.ix_(order, order)], cmap="viridis")
ax[0].set(xticks=range(6), xticklabels=np.array(sample_names)[order],
          yticks=range(6), yticklabels=np.array(sample_names)[order],
          title="Reordered distance matrix")
dendrogram(Z_link, labels=sample_names, ax=ax[1])
ax[1].set_title("Complete-linkage dendrogram")
plt.tight_layout()

```



Questions

1. How does a dendrogram encode multiple possible clusterings, and what does a horizontal cut at a chosen height mean?
2. Compare single, complete, and average linkage. How can the linkage choice change the tree or resulting groups?

7. *k*-means

Fit *k*-means for several values of *k* to the original toy data.

```

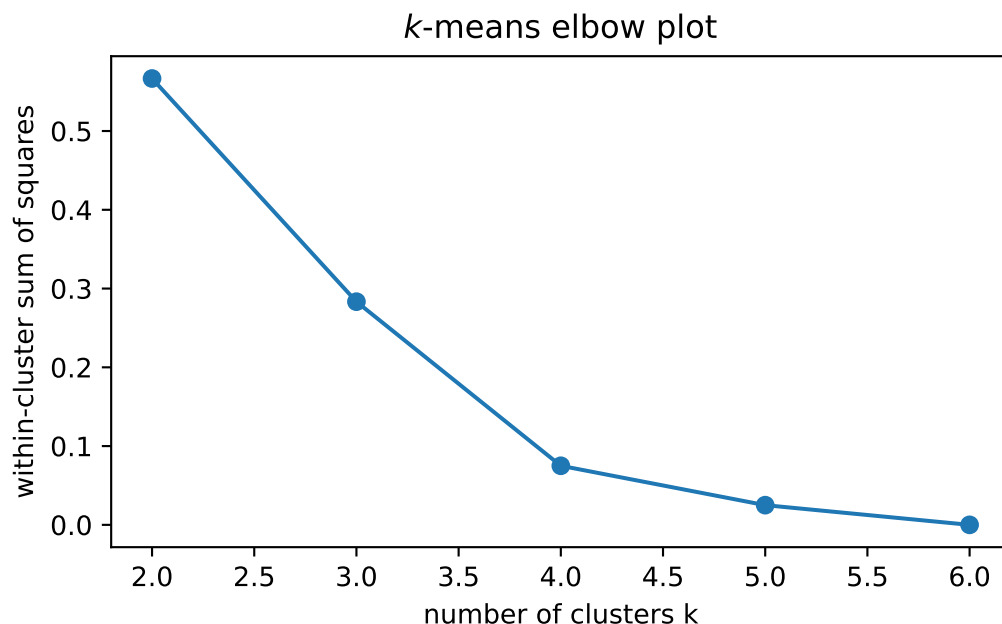
from sklearn.cluster import KMeans

candidate_k = range(2, 7)
inertia = []
for k in candidate_k:
    model = KMeans(n_clusters=k, n_init=20, random_state=375).fit(X_small)
    inertia.append(model.inertia_)

plt.plot(list(candidate_k), inertia, marker="o")
plt.xlabel("number of clusters k")
plt.ylabel("within-cluster sum of squares")

```

```
plt.title("$k$-means elbow plot")
plt.tight_layout()
```



Questions

1. What objective does k -means minimize? Describe its assignment and update steps.
2. What does the elbow plot suggest? Why is an elbow not proof that one k is correct? Run several random seeds and comment on stability.

8. Neighbor graphs and clustering evaluation

Construct a k -nearest-neighbor graph for the six samples. You may use `sklearn.neighbors.NearestNeighbors` or draw the graph by hand from the distance matrix.

Questions

1. What information is retained by a k NN graph, and what is lost compared with the full distance matrix? Why can graph-based clustering help when global Euclidean geometry is fragile?
2. Calculate silhouette scores for candidate k -means solutions on a larger simulated dataset. What do high and low scores indicate?
3. If curated cell-type labels are available, calculate ARI. When is ARI appropriate, and when is silhouette score appropriate? What evidence would make you trust a cluster as a biological subgroup?

Final synthesis

Choose one PCA result and one clustering result from this lab. Write a short paragraph addressing:

1. What scientific question were you asking, and what method did you choose?
2. What pattern did you observe, and what does the method preserve or optimize?
3. What uncertainty, tuning choice, or technical confounder could affect the pattern? What additional evidence would strengthen your conclusion?

Submission checklist

- a notebook or equivalent file containing your code
- responses to all questions in both the Monday and Wednesday parts
- labeled and interpretable figures
- numerical results with appropriate interpretation
- a final synthesis paragraph
- a note identifying collaborators, if you worked in a group